

Algorithm Lab (CS2271)

Assignment 6

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Q1. Problem Statement

Write a C program to represent and analyze two real-world networks using **graph data structures**, where one graph is **sparse** and the other is **dense**. Let the number of vertices be $n = 100$. For the sparse graph, take the number of edges as $e_s = 120$, such that

$$e_s \ll \frac{n(n-1)}{2},$$

and generate the edges **randomly**. Represent this sparse graph using an **adjacency list** for both **unweighted edges** (showing only connections) and **weighted edges** (assigning a random cost or distance to each edge).

For the dense graph, take the number of edges as $e_d = 4500$, such that

$$e_d \approx \frac{n(n-1)}{2},$$

and generate the edges **randomly**. Represent this dense graph using an **adjacency matrix** for both **unweighted and weighted edges**. The program should **automatically generate random vertex pairs and edge weights**, construct the corresponding graph representations, and **display the adjacency list and adjacency matrix**.

Q2. Problem Statement

Write a C program to connect 6 cities (C1–C6) with minimum cost using **Prim's Algorithm** and a **Min-Priority Queue**. The cost of connecting each pair of cities (in lakhs) is:

$$\begin{aligned}(C1, C2) &= 3, & (C1, C3) &= 1, & (C1, C4) &= 6, \\(C2, C3) &= 5, & (C2, C5) &= 3, & (C3, C4) &= 5, \\(C3, C6) &= 4, & (C4, C6) &= 2, & (C5, C6) &= 6\end{aligned}$$

Your program should:

1. Represent the graph using an **adjacency list**.
2. Implement **Prim's Algorithm** starting from vertex C1 using a **Min-Priority Queue**.
3. Display the edges in the **Minimum Spanning Tree (MST)**, the **total minimum cost**, and the **order of vertices added**.

Q3. Problem Statement

Write a C program to connect 6 cities (C1–C6) with minimum cost using **Kruskal's Algorithm**. The cost of connecting each pair of cities (in lakhs) is:

$$\begin{aligned}(C1, C2) &= 3, & (C1, C3) &= 1, & (C1, C4) &= 6, \\(C2, C3) &= 5, & (C2, C5) &= 3, & (C3, C4) &= 5, \\(C3, C6) &= 4, & (C4, C6) &= 2, & (C5, C6) &= 6\end{aligned}$$

Your program should:

1. Represent the graph using an **edge list**.
2. Sort all edges in ascending order and construct the **Minimum Spanning Tree (MST)** using **Union-Find (Disjoint Set)** to prevent cycles.
3. Display the edges included in the MST, the **total minimum cost**, and the **order in which edges are added**.

Q4. Problem Statement

Write a C program to compute the **Minimum Spanning Tree (MST)** of a large-scale **smart city power distribution network** consisting of two independent weighted graphs $G_1(V_1, E_1)$ and $G_2(V_2, E_2)$, interconnected by a **single high-capacity transmission edge** having the **maximum weight among all edges**. This high-weight edge must always be included in the final MST. Apply **Kruskal's algorithm** on graph G_1 , considering it as a **sparse graph**, and apply **Prim's algorithm** on graph G_2 , considering it as a **dense graph**. After computing the MSTs of both graphs separately, merge them using the **mandatory high-weight edge** to construct the **global MST** of the combined network. The program should take the number of vertices, edges, and weighted edge lists as input, compute the **final MST**, display the **selected edges**, and output the **total cost** of the spanning tree.